

7 marks (4)

Fuzzy logic
family → 1

Bipolar logic families

The main elements of a bipolar IC are resistors, diodes (capacitors) & transistors. They're two types of operation in bipolar ICs:

① Saturated ② Non saturated.

Saturated Logic: The transistors in the IC are driven to saturation. The saturated logic bipolar logic families are: ① RTL ② DCTL [Direct coupled transistor logic]

③ I²L [integrated-injection logic] ④ DTL

⑤ HTL [High threshold logic] ⑥ TTL

Non Saturated: The transistors are not driven into saturation. The non-saturated bipolar logic families are: ① Schottky TTL

② Emitter Coupled Logic.

Unipolar: ^{MOS logic family} ① PMOS ② NMOS ③ CMOS.

① in PMOS: while in PMOS only p-channel MOSFETs are used

② NMOS: only n-channel MOSFETs are used

③ CMOS: both p- & n-channel MOSFETs are employed & are fabricated on same silicon chip.

Digital Logic IC Families

Logic Family: A group of IC's with same logic level & supply voltage are fabricated together to perform various logical function and referred as logic family.

Type of logic level:

- ① Positive
- ② Negative

Type of semiconductor device:

① Unipolar

↓
charge carrier
(hole or electron)

② Bipolar

↓
charge carriers
(hole and electron)

- saturated
 - ① Use semicon device
 - ② On or off condition
- non saturated
 - ① active region
 - ② ECL
 - ③ TTL

Digital logic family & type:

① Bipolar
⇒ electron & hole
charge carriers

② Unipolar
⇒ electron or
hole charge
carrier

③ BiCMOS
⇒ I/P mos
⇒ O/P mos

Bipolar

Saturated

- Use semiconductor device
- On state, OFF state
- ↓
Saturated region
- RTL, TTL, DTL

Non-saturated

- Active Region
- ECL

Advantage

- Reduce size of the system
- Reduce power consumption
- Reduce cost
- Reliable

Disadvantage

- It can't handle large power
- Suitable for low power consumption application.

Characteristics of Digital IC's:

Supply Seq. of operation

① Propagation Delay: Time a signal takes to propagate through input to output.

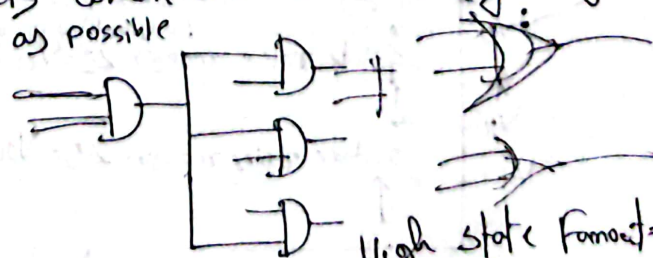
② Threshold voltage: At what voltage transition happen from logic low to logic high.

③ Power dissipation: How much voltage dissipated as heat. This is the amount of power dissipated in an IC. Determined by $V_{cc} \times I_{cc}$. I_{cc} is the avg. value of $I_{cc}(0)$ & $I_{cc}(1)$.

④ Figure of Merit: Merit: Product of Propagation delay $\times$ Power dissipation. $\text{Figure of merit} = \frac{1}{\text{Propagation delay} \times \text{Power dissipation}}$

It is defined as product of speed & power. $(\text{ns} \times \text{mW}) = \text{pJ}$

⑤ Operation temperature: Fan out: Number of similar gates which can be driven by a gate, should be as high as possible.



$$\text{High state fanout} = \frac{I_{OH}}{I_{IH}}$$

$$\text{Low state fanout} = \frac{I_{OL}}{I_{IL}}$$

imp Fan in: number of input of a gate that it can handle without hampering its normal operation.

imp Noise Tolerance: Circuit ability to tolerate noise. Stray electric & magnetic fields may induce unwanted voltages, known as noise on the connecting wires between logic circuits.

imp Operation Temperature:

① Commercial application: 0°C to 70°C

② Industrial application: 0°C to 25°C

③ Military: -55°C to 125°C

imp Speed power product: Measuring & comparing the overall performance of IC family. It is the product of propagation delay in second & power dissipation in watts per second.
$$\text{Speed power product} = \text{propagation delay (s)} \times \text{power dissipation (W)}$$

Noise tolerance: (Circuit ability to tolerate noise signal is referred to as noise immunity) (a quantitative measure of which is called noise margin)

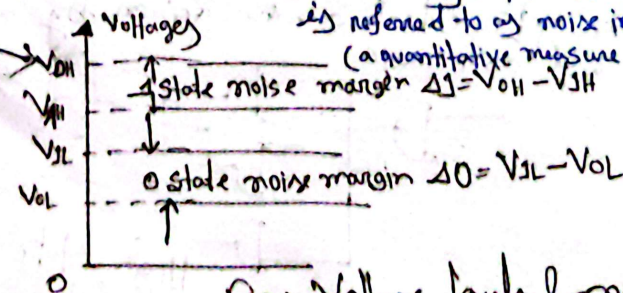
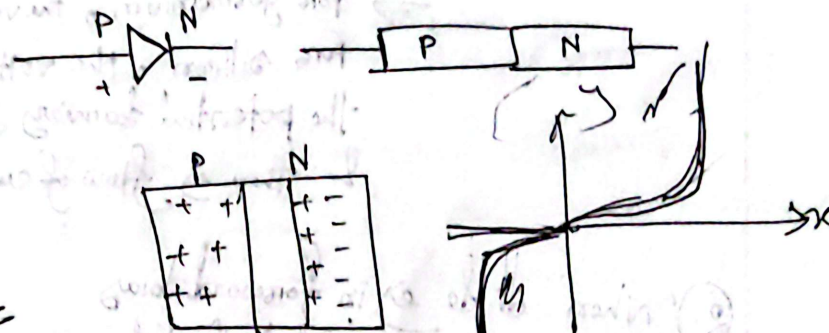


Fig: Voltage levels & noise margin of ICs

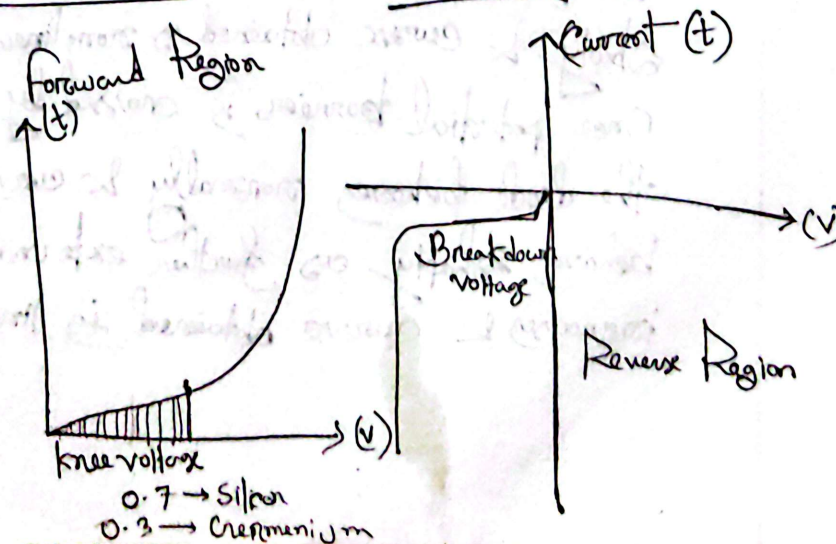
(1) diode (6)

A diode is a semiconductor device that act as an one way switch for current.

It allows current to flow easily in one direction.
 & restricted current flow in opposite direction.



V-I characteristic: Voltage x , current y



~~Switching~~

- ① In forward bias $\rightarrow$ p-type is connected to positive terminal of battery
 $\rightarrow$ n-type is connected to negative terminal of battery.
 $\Rightarrow$ For germanium, the voltage 0.3V
For silicon, the voltage 0.7V,
the potential barrier decrease
& there is flow of current.

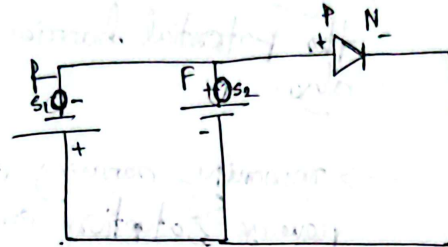
- ② When diode is in forward bias
 $\rightarrow$ as the voltage applied to the diode is overcoming the potential barrier, the current increases sharply & curve obtained is non-linear.
Once potential barrier is crossed by diode, the diode behaves normally & curve behaves sharply as further external voltage increases & curve obtained is linear.

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- ③ When ~~PN diode~~ ^{When P-N} Reverse diode bias, the
- this results in an increase in the potential barrier & resistance also.
 - minority carriers create reverse saturation current flow in beginning.

④ If applied voltage increases rapidly, there is increased kinetic energy due to minority charge carriers which affect majority charges. In this stage the diode breaks down. On the voltage is called breakdown voltage.

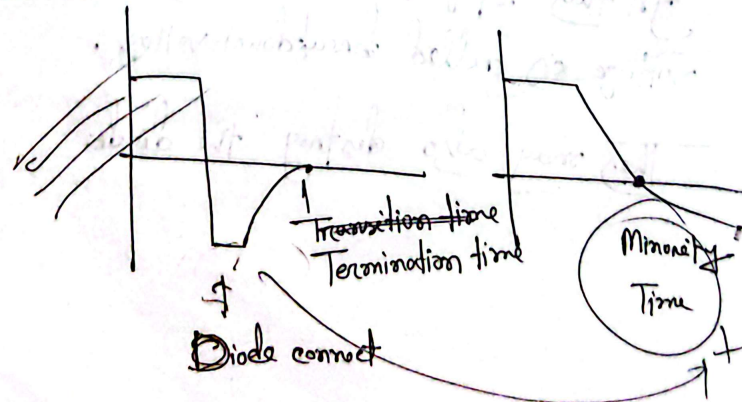
They may also destroy the diode.

Switching characteristics of diode:



When we put a diode from forward bias to reverse bias, it shows certain characteristics.

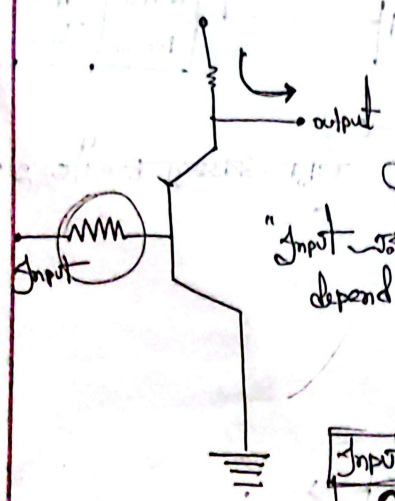
Forward to reverse:



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RTL (Register Transistor Logic)

Not gate

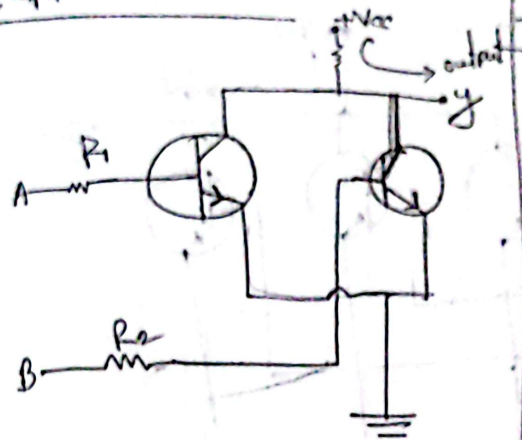


"Input is ^{either} Ground or output depend on it."

Input	Output
0	1
1	0

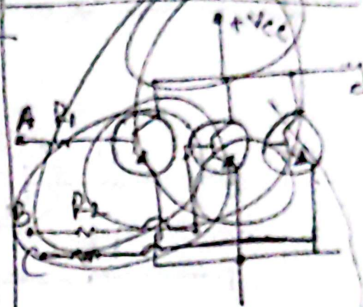
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2 input NOR Gate:



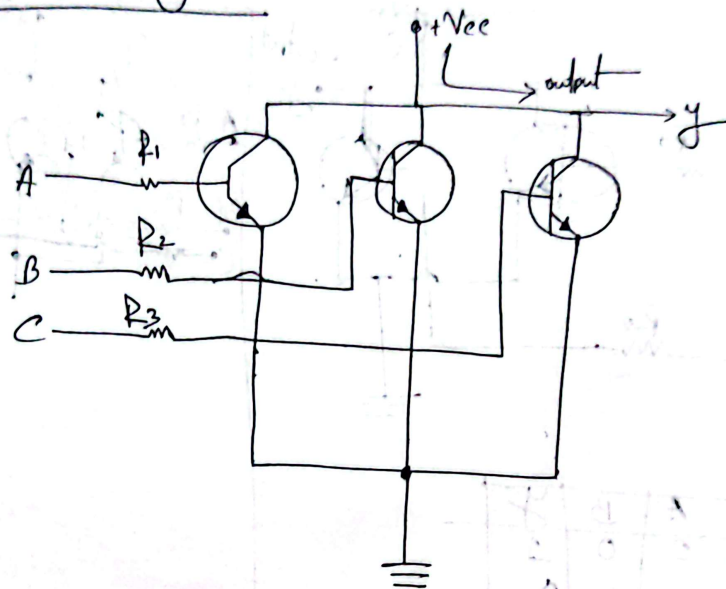
A	B	y
0	0	1
0	1	0
1	0	0
1	1	0

3 input NOR Gate:



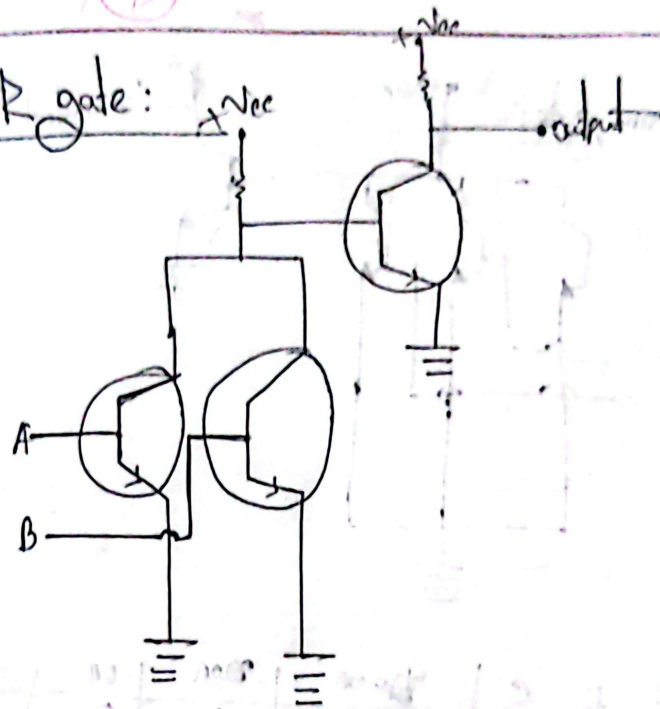
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3 input Non gate



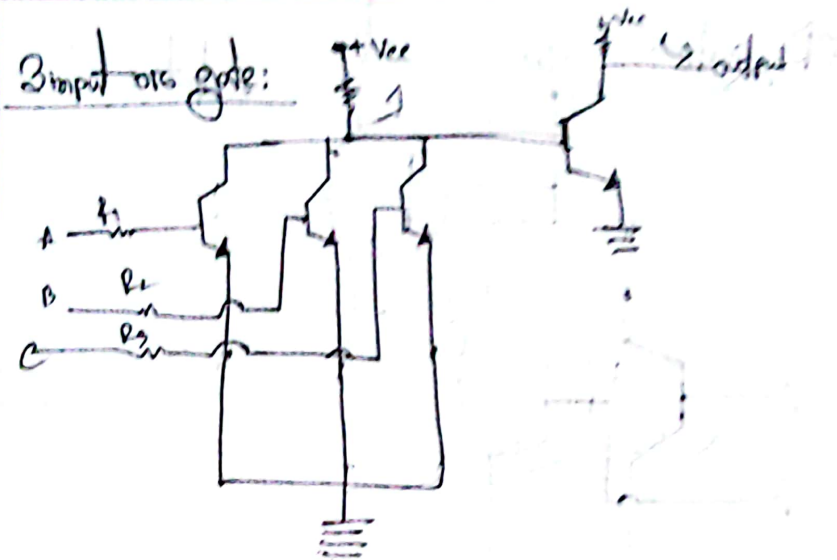
A	B	C	y
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

OR gate:



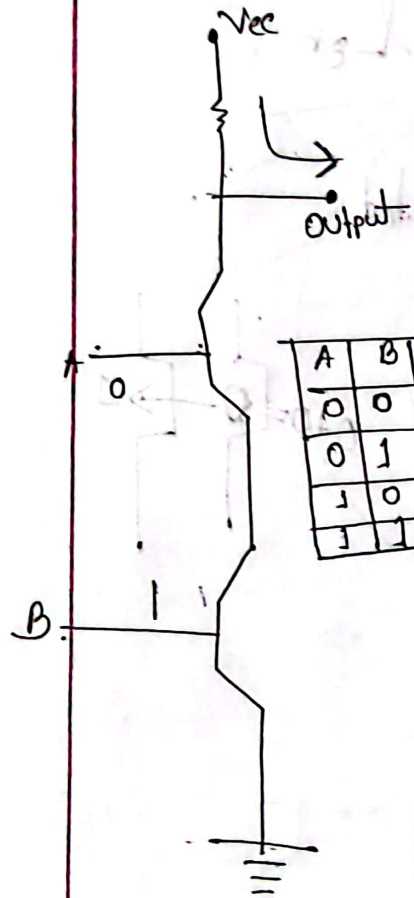
A	B	Not	Of	
0	0	1	0	→ voltage 1, no huge power 1
0	1	0	1	→ voltage 1, may not gate 2
1	0	0	1	→ voltage 0, may not gate 2
1	1	0	1	→ voltage 0, may not gate 2

2 input or gate.



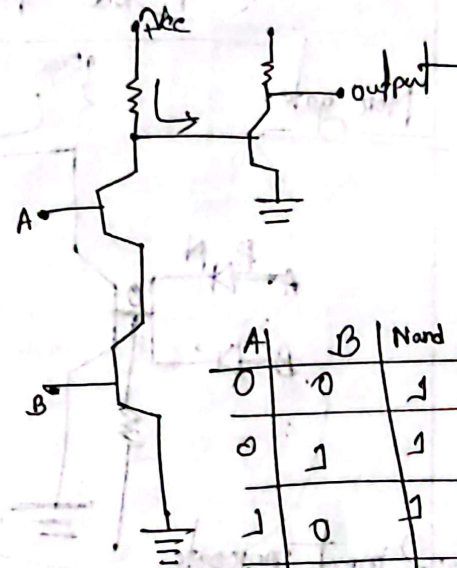
A	B	C	Y	non	on
0	0	0	0	1	0
0	0	1	1	0	1
0	1	0	1	0	1
0	1	1	1	0	1
1	0	0	1	0	1
1	0	1	1	0	1
1	1	0	1	0	1
1	1	1	1	0	1

NAND gate:



A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

AND gate:

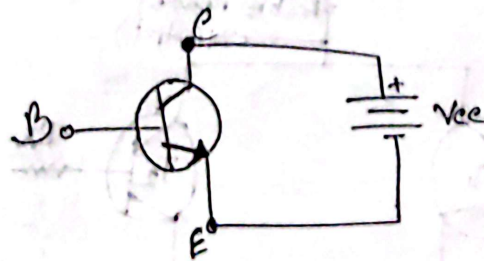


A	B	Nand	And
0	0	1	0
0	1	1	0
1	0	1	0
1	1	0	1

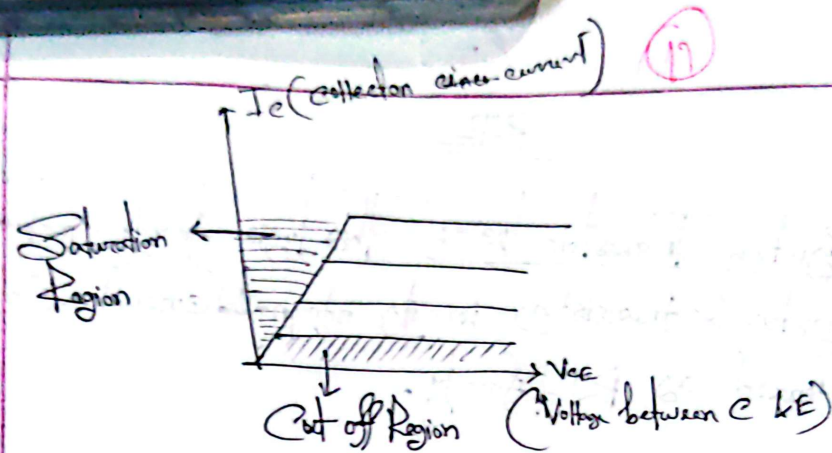
RTL

Register Transistor logic: This logic family includes register & transistors in its integrated circuit. So it is referred as RTL family.

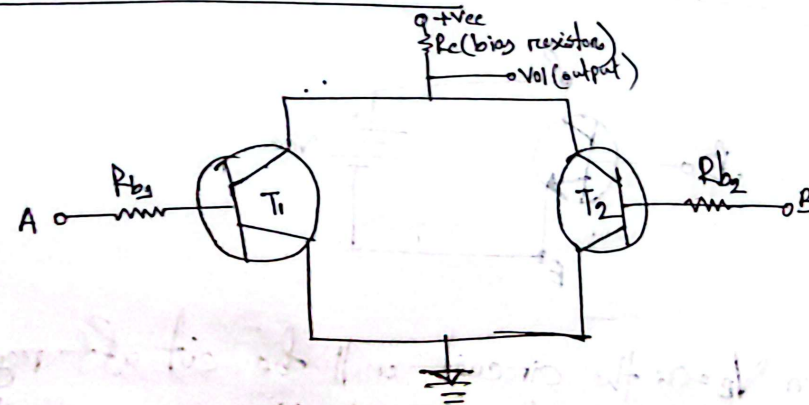
* Here we use transistor in saturation & cut off regions.



- ① When $V_B = 0$: the circuit will be cut off region, so, C to E terminal will act like open circuit.
- ② When, $V_B = \text{high or } V_{cc}$, the circuit will be on saturation region, so, current'll flow from C to E.
- ③ When the transistor is in saturation mode the potential voltage drop will be 0.2V, $\therefore V_{CE} = 0.2V$



NOR gate using RTL: (working principle)



- ① Both 0 means they're in cut off region, output will be 1
- ② One 0, & another 1 transistors will be in ~~cut off~~ cut off region & another will be at Saturation.

As current flow follows shortest path rule, it'll go through saturation region into Vce, where voltage will be 0.2V considered low or 0.

③ When applied voltage is 1, the both transistor will be saturation. So output voltage will be low.

(~~Ques~~ working principle for NAND, OR, AND explain)

Limitations:

① When the transistor is switched on, the power dissipation increases as current flows through base & collector (high power dissipation)

② Poor noise margin.

③ Poor fan out

④ Propagation delay is more

⑤ Can not operate at speed above 4MHz.

power
dissipation

DTL Diode Transistor Logic

Nor gate:

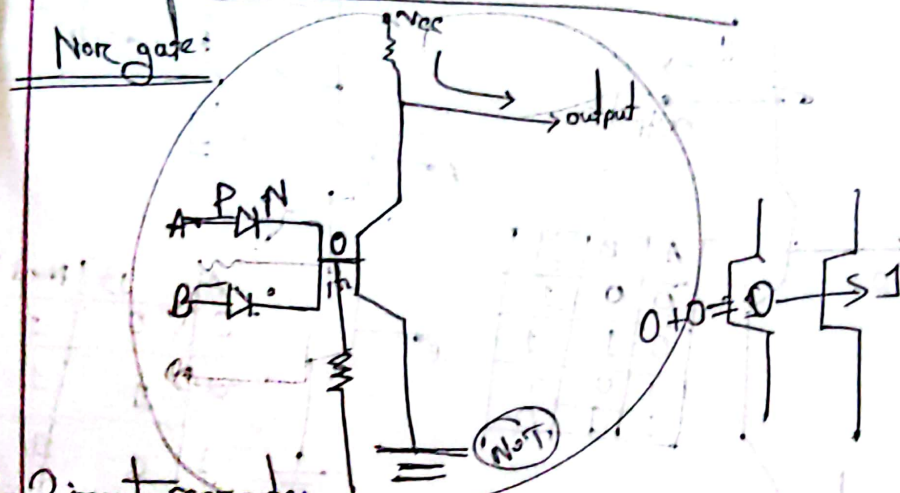
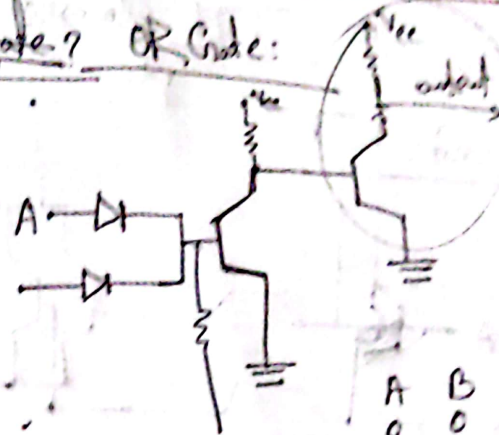


Fig: 2 input nor gate:

A	B	NOR
0	0	1
0	1	0
1	0	0
1	1	0

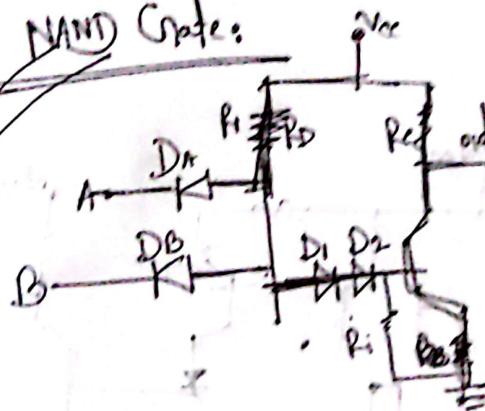
3 input nand gate? OR Gate:

TTL

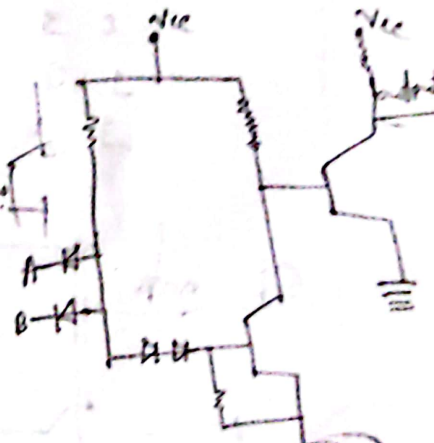


A	B	C
0	0	0
0	1	1
1	0	1
1	1	1

NAND Gate:



A	B	A NAND B
0	0	1
1	0	1
0	1	1
1	1	0



A	B	A NAND B
0	0	1
1	0	1
0	1	1
1	1	0

DTL

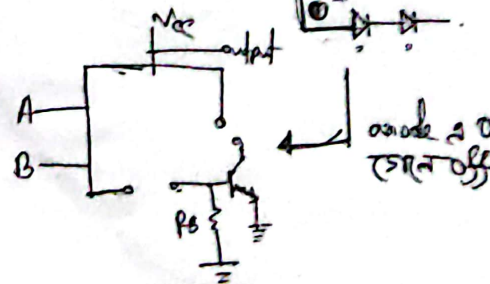
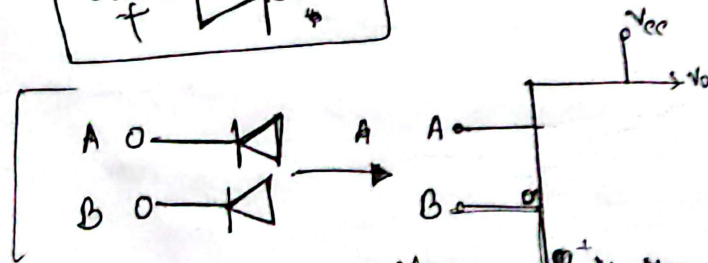
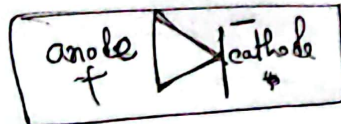
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Q1 Draw the circuit diagram of 3 input DTL-NAND Gate. Calculate f_{out} , noise margin & average power if the parameters are considered as voltage across a conducting diode = 0.7V, cut in voltage of diode = 0.6V, cut in voltage of transistor = 0.5V, $V_{BE, sat} = 0.8$; $V_{CE, sat} = 0.2$; $\beta_{FE} = 20$. f.

for 0, 0 - input

Split DTL

Q1 Cathode of diode must be low. then it will be short circuit, (forward bias) (register as 3 input plus diode)



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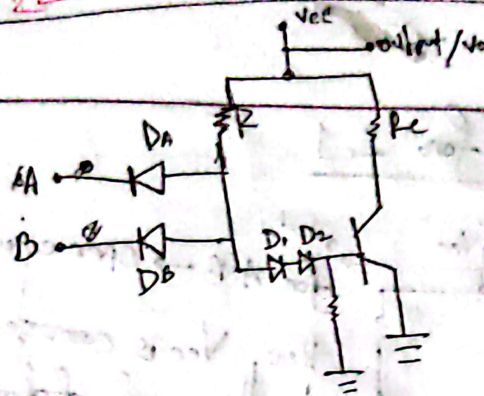
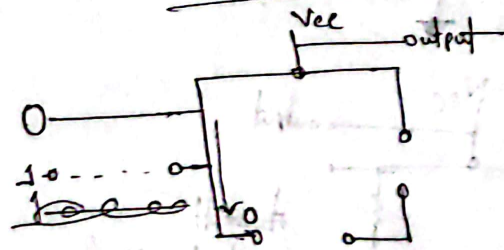
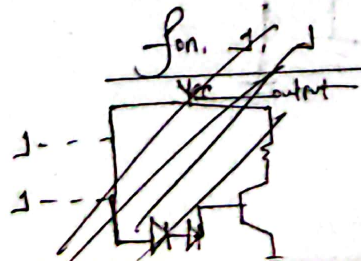


Fig: 2 input NAND DTL NAND.

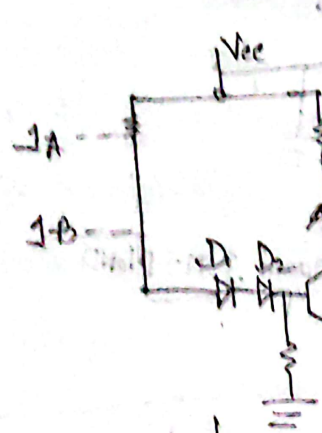
for 1, 0 - input



for 0, 1 - input - 2nd

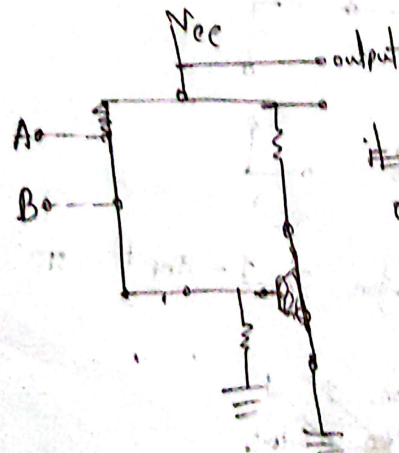


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 Ex 1.1



for A, B $\rightarrow$ Open circuit
 Here V_{cc} is connected
 directly anode of diode D_1
 D_2

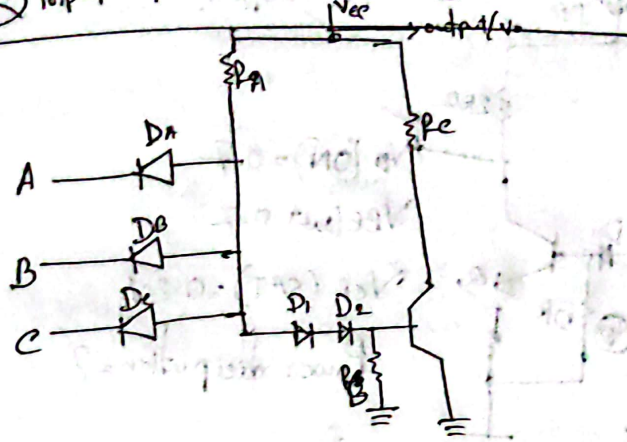
V_{cc} on high voltage
 connected to anode
 then there'll create
 short circuit

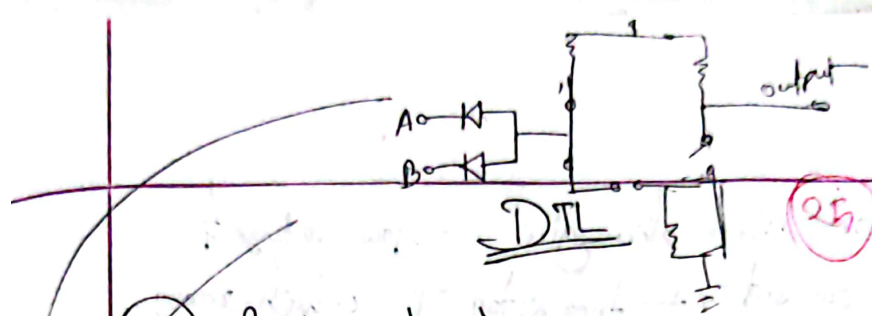


it will
 output will be 0.

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Ans: 3 input NAND DTL NAND gate;





Diode transistor logic:

Nand gate implementation: (exercise)

0	0	1
1	0	1
0	1	1
1	1	0

① When voltage applied with both low signal the D_A & D_B will both act as forward bias. Whereas D_1 & D_2 will act as reverse bias.

② As D_1, D_2 are working as reverse bias, the bias current of transistor T is low, T will be in cut off mode (open circuit). So output will be V_{cc} . ~~V_{cc}~~ $V_{out} = V_{cc}$.

③ Again when apply high voltage, to one input & low voltage to another, we'll get one of the diode D_A or D_B in forward and other will be reverse bias. D_1 & D_2 will work on active

reason, which also give a low voltage to base current of transistor T. which means it's on the cut off region. So output will be V_{cc} or high voltage.

④ Both inputs gets high voltage, input D_A & D_B works on reverse bias

⑤ D_1 , D_2 works on forward bias. For this reason, the base current of transistor T get high voltage. So it'll be on Saturation mode & works as short circuit. Then output becomes 0.2V on low voltage which is considered as logic 0.

(27)

OP ~~NOT~~ CMP OP

1 Disadvantage of DTL:

- ① Low speed
- ② Power dissipation.
- ② high noise immunity

Why we use D_1 & D_2 :

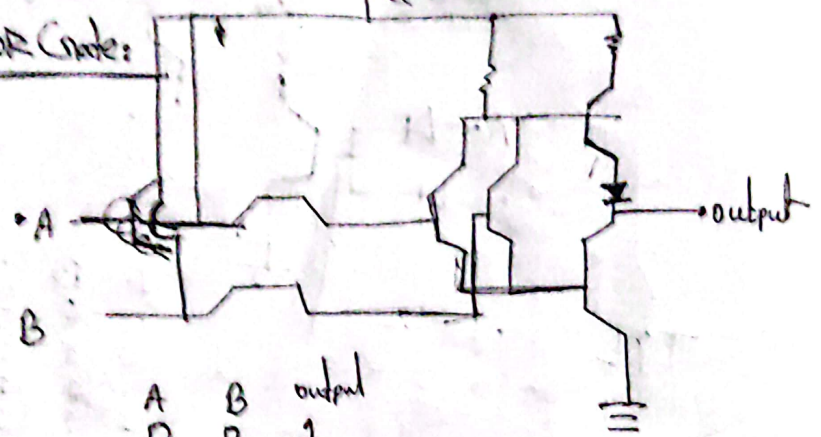
- ① Noise Margin getting improved than RTL
- ② improving fan out

Fan out $\rightarrow$ Low
Not $\rightarrow$ High

Q2

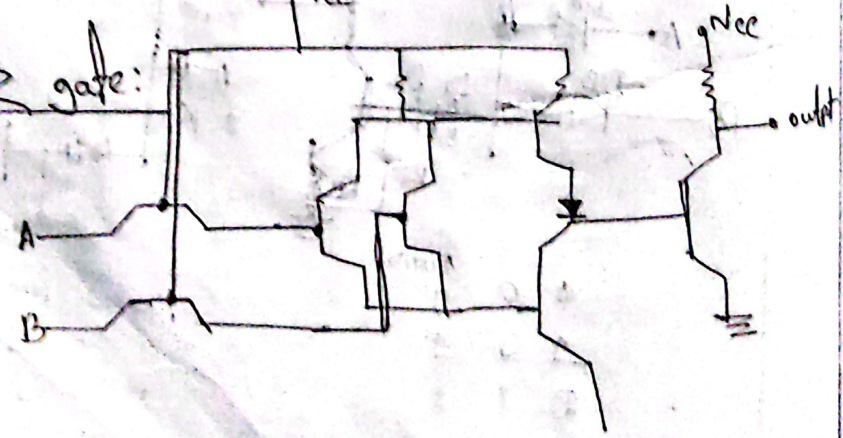
TTL (transistor transistor)

NOR Gate:



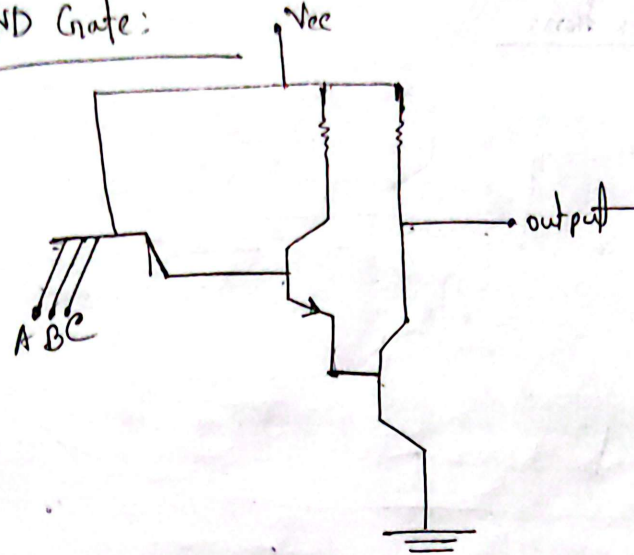
A	B	output
0	0	1
0	1	0
1	0	0
1	1	0

OR gate:

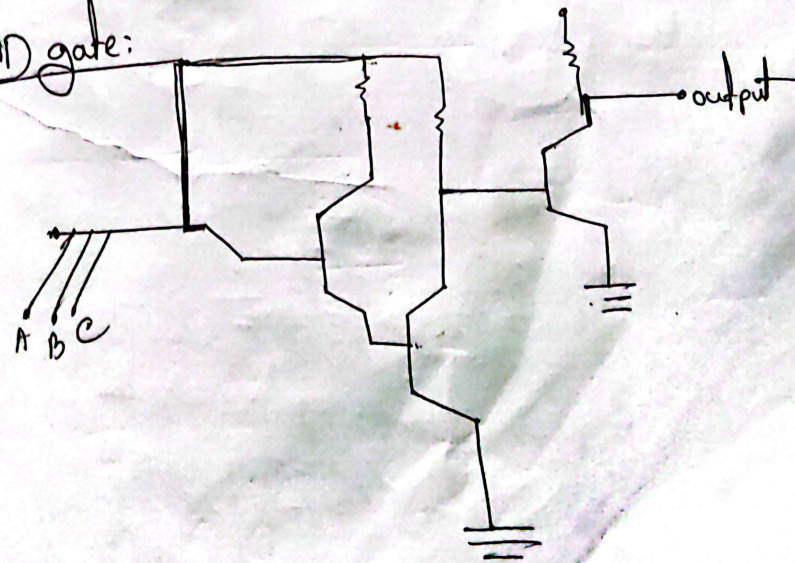


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NAND Gate:

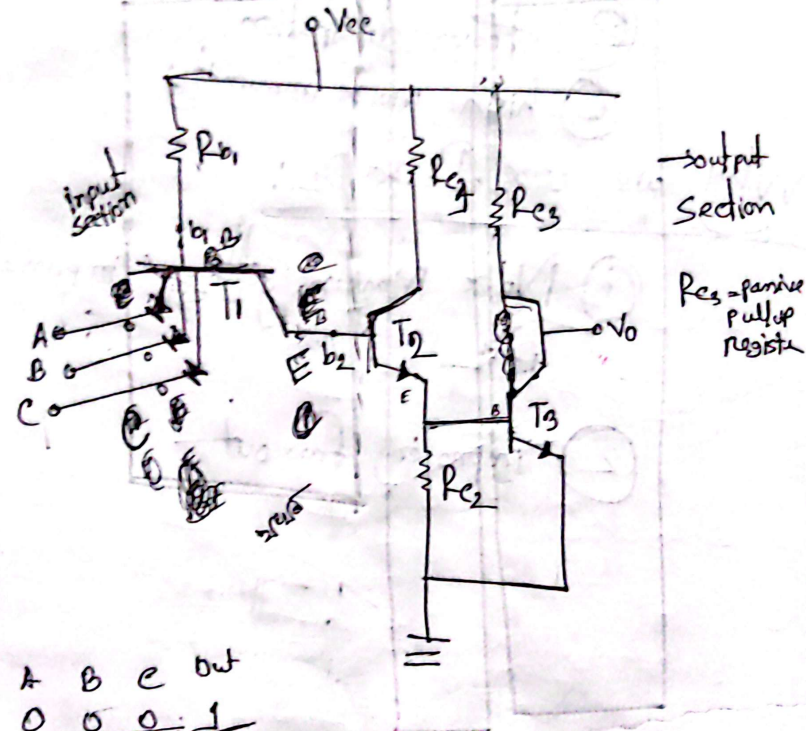


AND gate:



TTL

NAND gate:



A	B	C	out
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

phase
splitting
section

31

① When any of input is low means logic '0'.
The base emitter for T_1 will act as forward and base collector works as reverse bias for this reason, base input for T_2 will be low.
So it will be cut off region.

② T_2 is in cut off region, T_3 will be on cut off region. So output will be high on logic 1 on Vcc.

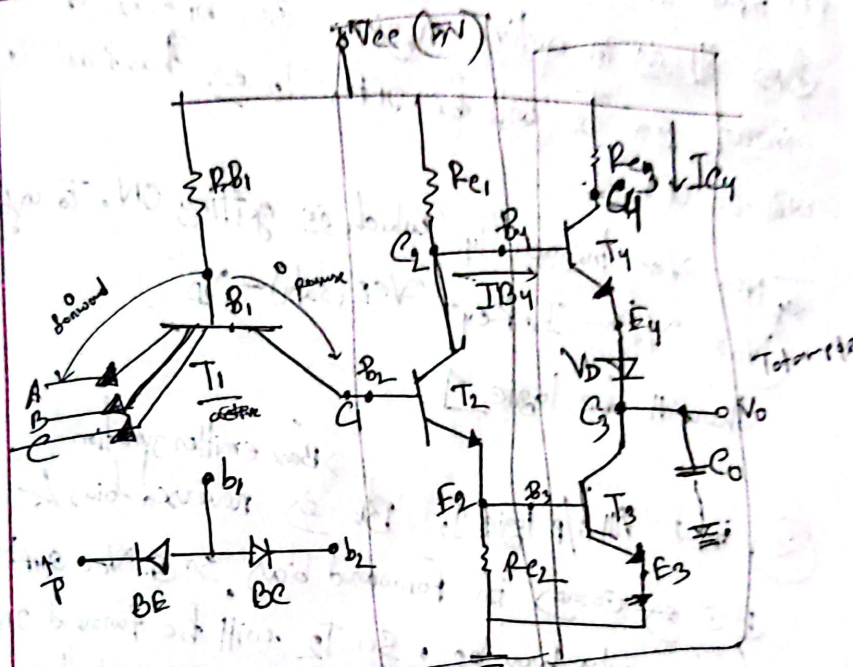
② In the other hand, when the inputs are all 1 on high, the base emitter for T_1 will work as reverse bias & base collector as forward bias.

For this reason base input T_2 will become high & T_2 act as in an saturation mode, As T_2 is on so there will also be high

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input at T_3 base. Then T_3 will also act on saturation region. Means it will act as a short circuit. So the output voltage becomes Low on logic 0.

(33) NAND gate implementation with TTL (totem pole).



V_o for all 0 on T_4 only $0 = V_{CC} - I_{C4}R_{C4} - V_{CE4} - V_D$ (Solution)

To turn on T_4 , when all input is 0, we need $0.7 + 0.7 = 1.4V$, But as T_2 & T_3 are in saturated mode. The voltage supply for T_4 by will become $V_{CE2} + V_{BE3} = 0.2 + 0.7 = 0.9V$

working principle

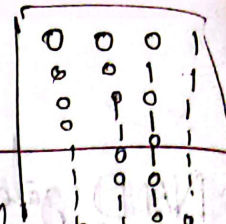
① input low, ~~base~~ BE is forward & BE is reverse bias. T_1 is in active region. So, b_1 & b_2 open circuit so T_2 will be OFF, T_3 based on T_2 will be OFF.

Now Vcc trigger T_1 which is getting ON. T_2 will be ON.

$$V_o = V_{cc} - I_{Ceq} R_{eq} - V_{CE(sat)} - V_D$$

 there'll be logic 1

② input All 1/p logic 1, ^{Base emitter junction} BE is reversed bias & BE is comes in forward bias. So T_1 Vcc supply, draw a supply base b_2 , so T_2 will be turned ON. T_3 will also Vcc supply draw a supply at base b_3 & T_3 will also be turned ON. base terminal
 Justify whether T_4 is ON or OFF. V_{be} will become $V_{BE1} + V_{BE3} =$



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TTL Limitation:

- ① Power dissipation
- ② limits no of I/P can be connected
- ③ relatively high o/p resistance at o/p logical 1.

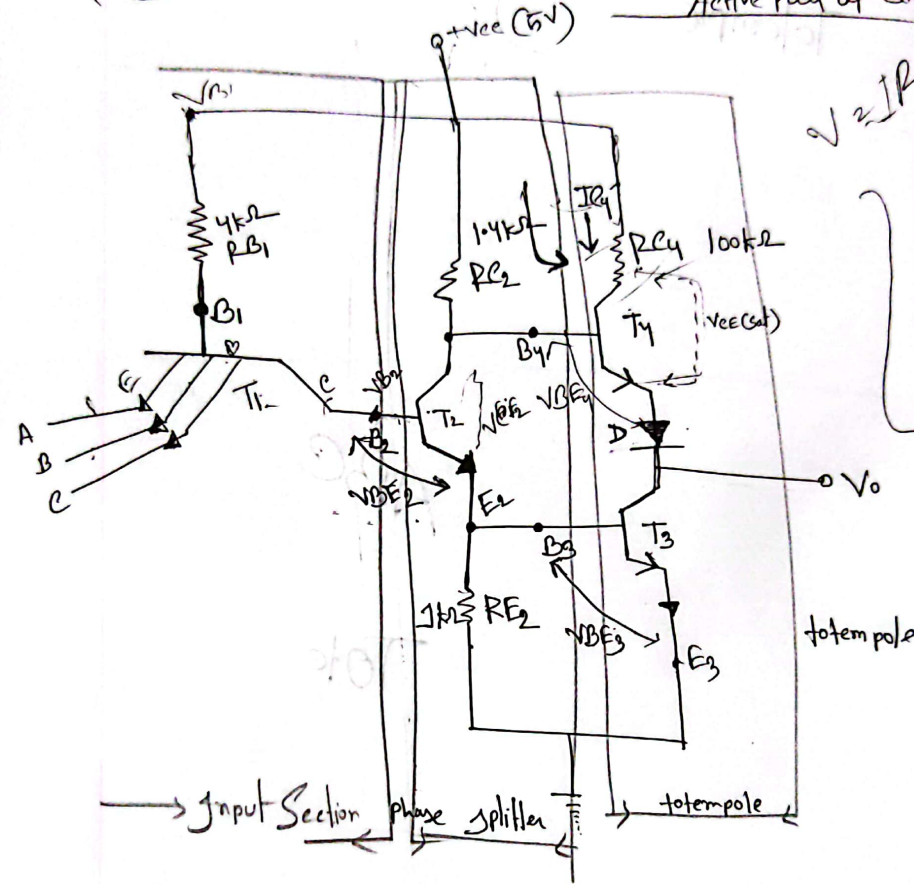
Advantages of TTL:

- ① Higher speed
- ② Higher noise immunity
- ③ High output current
- ④ Proven technology
- ⑤ Lower power consumption in idle state.

(1)

36 Totem pole

Totem pole on Active Pull up circuit



It carries active pull up for capacity load. Active pull up transistor off state is shown, T4 a low resistance offer, low impedance offer, T3 ON state is shown, T4 a high impedance offer, T3 a low impedance offer. The totem pole is shown.

Low 0

T_2 off state means open circuit.

$\Rightarrow V_{EE}$ provides sufficient voltage to turn on T_4 .

$T_4 = \text{ON}$,

$$V_{out} = V_{EE} - R_{E4}(I_{E4}) - V_{BE4}(\text{sat}) - V_{D0.7}$$

(5 ---) (0.2V) FB

$V_o = \text{HIGH}$.

$$V_{B1} = 1.3 + 0.65 \\ = 1.95V$$

When input High:

① $B_E \rightarrow$ Reverse bias Forward

② $B_E \rightarrow$ Forward bias Reverse

③ $T_1 \rightarrow$ Forward region.

④ $T_2 \rightarrow \text{ON}$, $T_3 \rightarrow \text{OFF}$, [B_3 has sufficient voltage to make T_3 ON]

⑤ T_4 ON means, short circuit (0.2V)

$$\begin{aligned} V_{B4} &= V_{E4}(\text{sat}) + V_{BE4} \\ &= 0.2 + 0.7 \\ &= 0.9V(\text{at } B_4) \end{aligned}$$

⑦ To turn $T_4 = V_{BE4} + V_D = 0.65 + 0.65 = 1.31$

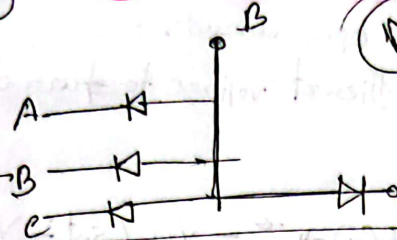
$T_4 = \text{OFF}$, because cut OFF. because $0.9 < 1.31$.

T_4 requires 1.31V but it has 0.9V,

(2)

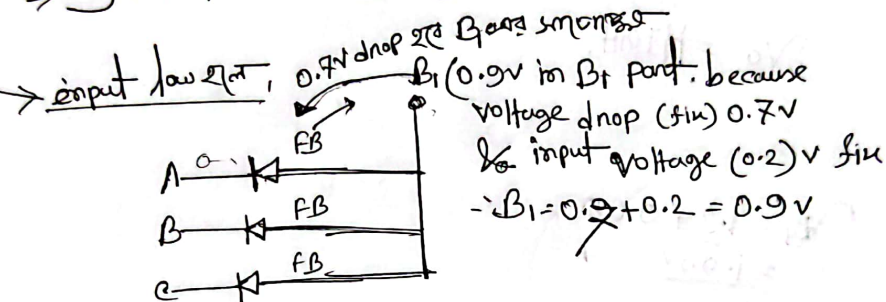
(38)

(N)



$V_D = 0.7$
 $V_{BE} = 0.8$
 $V_{CE} = 0.2$
 $V_{CC} = 5V$

⇒ Initial low input supply. Let $0.2V \therefore V_{CE(sat)} 0.2V$



⇒ ~~off~~ voltage required $V_{B2} = V_{BE2} + V_{BE3}$
 $= 0.65 + 0.65$ (off state)
 $= 1.3V$ (required)

But given in B_1 is $0.9V$. So T_2 will be it's not sufficient $B_1 < V_{B2}$. So, T_2 will be in off region & so T_3 will be on off region.

I/P low: $T_1 = \text{active region}$ cause BE is forward & BC is in reverse bias. b_1 & b_2 open circuit
 $T_2 = \text{OFF}$. Emitter current not flow ^{of T_2} so we don't get ~~off~~ voltage on base of T_3 .
 $T_3 = \text{OFF}$

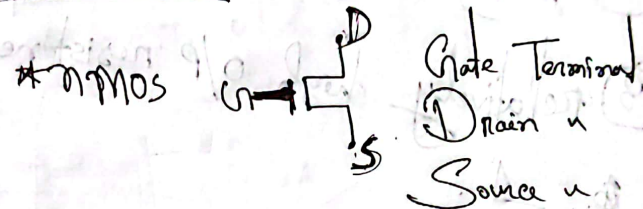
CMOS

CMOS

Complimentary
(Complement Metal Oxide
Semiconductor)

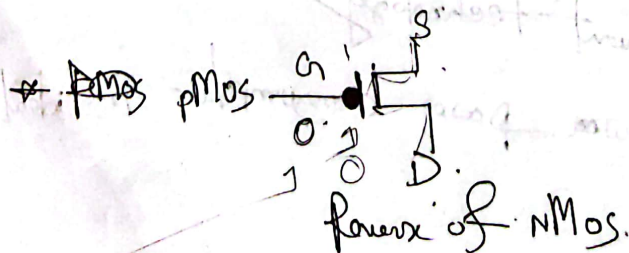
→ used to perform logical functions.
CMOS: Basically having two types transistor

nMOS & PMOS



① If $G=0$; the nMOS will at off state
So Drain to Source will act
like open circuit.

② If $G=1$; the nMOS will at 'ON'
State, short circuit.



(19) CMOS are used as a logic gate, then they're used as a switch. So in both, NMOS & PMOS transistor, the voltage which applied between gate & source terminal act as control voltage.

CMOS:

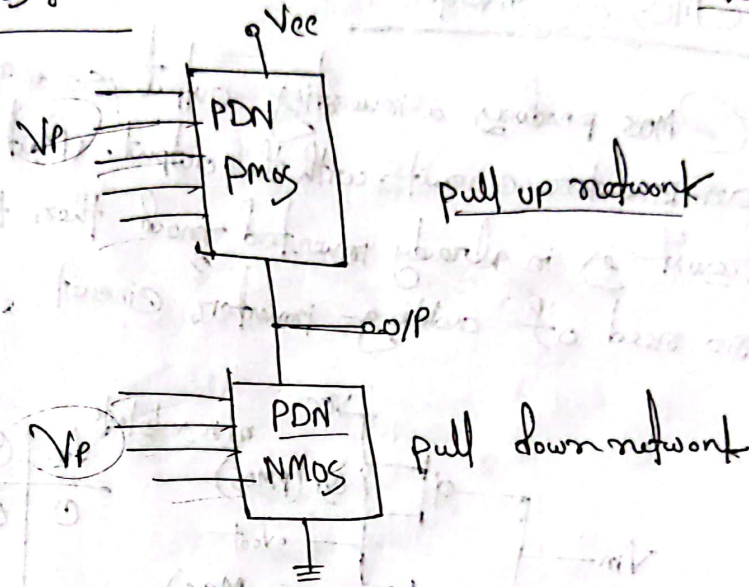


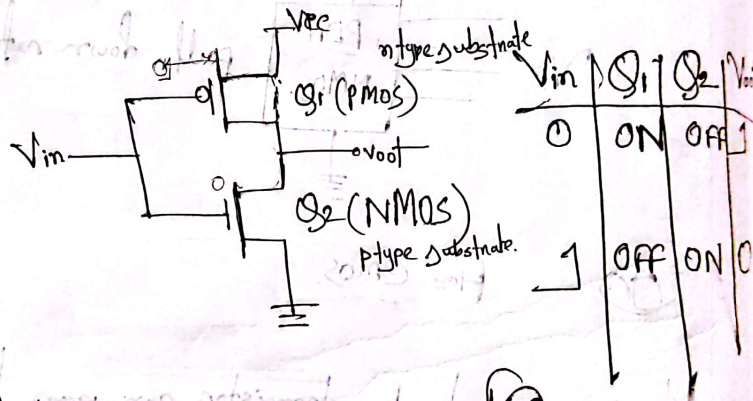
Fig: CMOS.

* In PMOS network transistor can pass logic high easily

In NMOS → logic Low

CMOS as Inverter:

CMOS produces a inverting output so ~~we~~ add an inverting circuit with the output. But if the result is in already inverted mode then there will no need of adding inverting circuit.

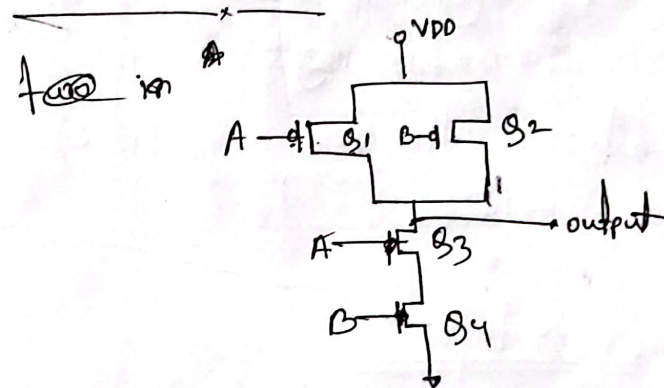


Advantages:

- ① Circuit is simple.
- ② Suitable for LSI
- ③ Power Usage less
- ④ Work on 15 to 20V
- ⑤ Fan out is 50
- ⑥ Noise margin low
- ⑦ Propagation is more than t_p
Propagation dissipation 0.01 mW.

Dis:

- ① Cost more than TTL
- ② Propagation delay is more
- ③ Stored in special conducting form.



input		P.Mos		N.Mos		output
0	0	Q ₁	Q ₂	Q ₃	Q ₄	1
0	1	ON	OFF	OFF	OFF	1
1	0	OFF	ON	ON	OFF	1
1	1	OFF	OFF	ON	ON	0

Fig: 2 input nand get using CMOS/NOR

1) ~~How~~ Explain how transistors are used as inverters? (3)

1) Draw the circuit of CMOS switch. (2018-2019) (2)

2) Show the working principle & applications of CMOS inverter. (2020) July-Dec (4) (Jul-Dec 2021)

Ans:
Ans:

Ans to the ques no-1 marks (3)

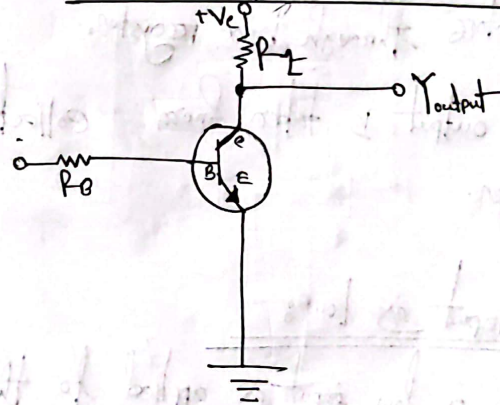


Fig: Transistor inverter.

A transistor can be used as inverter (NOT gate) in digital logic circuit by utilizing its switching property.

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① Working:

- ① The collector of transistor is connected to positive voltage (V_{cc}) through load resistor.
- ② The emitter is connected to ground.
- ③ The input signal is applied to the base of the transistor through base resistor.
- ④ The output is taken from collector of transistor.

② When input is Low:

- ① When a low input is applied to the base of transistor, the transistor remains in the cut-off region.
- ② No current flows through the transistor, it behaves like a switch open switch.
- ③ As a result the voltage at collector is pulled to V_{cc} .

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(4) So, the output is High when input is Low.

When input is High (1):

- ① When a high input voltage is applied to the base, the transistor enters to the saturation region.
- ② Transistor behaves like closed switch, allowing current flow from collector to emitter.
- ③ This causes the voltage at collector to drop to ground potential (0V).
- ④ output is low, when input is high.

input (Base)	Output (collector)
0 (Low)	1 (high)
1 (High)	0 (Low)

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CMOS Inverter Working Principle:

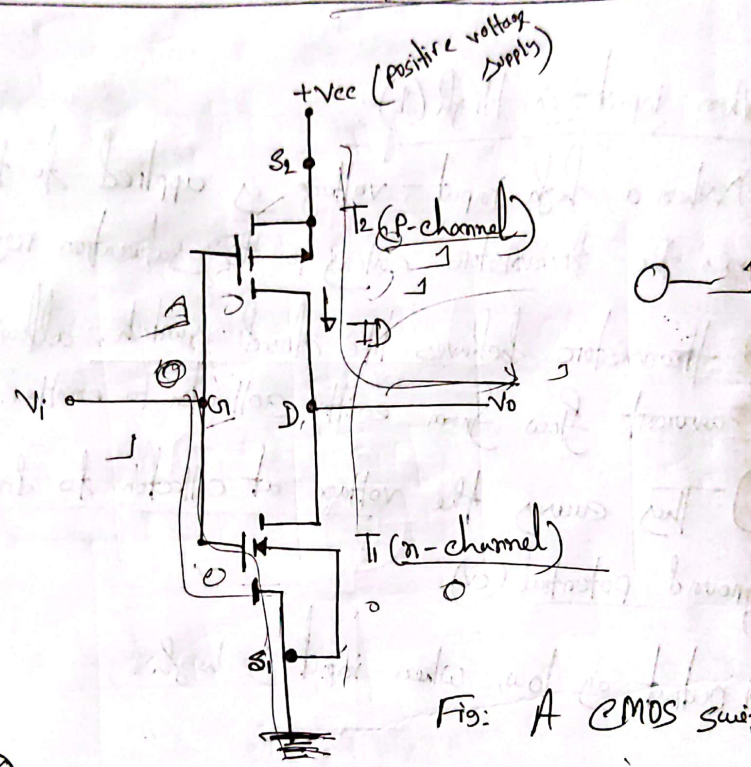


Fig: A CMOS switch

① Complementary MOSFET (CMOS) is obtained by connecting a p-channel & n-channel MOSFET in series with drains tied together & output is taken at the common drain.

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Working Principles:

The CMOS inverter operates in two stages:

(a) When the input is LOW (0V):

- (i) 0V at gate & Vcc at source: The PMOS transistor turns ON because the gate-source voltage of transistor is negative.
- (ii) The NMOS transistor turns OFF because its gate-source voltage is zero.
- (iii) The PMOS transistor pulls the output ^{to Vcc} ~~High~~ (Vcc), so the output is HIGH.
- (iv) When $V_i = V_{cc}$, T_2 turns ON & T_1 turns OFF.
- (v) $V_o \approx 0V$, transistor connected in series the current I_D is very small.

(b) When the output is HIGH (1V):

- (i) The PMOS transistor turns OFF because gate-source voltage is zero (both Vcc & gate).

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② NMOS transistor turn ON, because its gate source voltage is positive.

③ Pull output to ground (0V), so output is Low

Input (V_{in})	PMOS	NMOS	Output
Low (0V)	ON	OFF	HIGH (V_{DD})
HIGH (V_{DD})	OFF	ON	LOW (0V)

* Power consumption: Power consumption is very low little because only one transistor is on at a time.

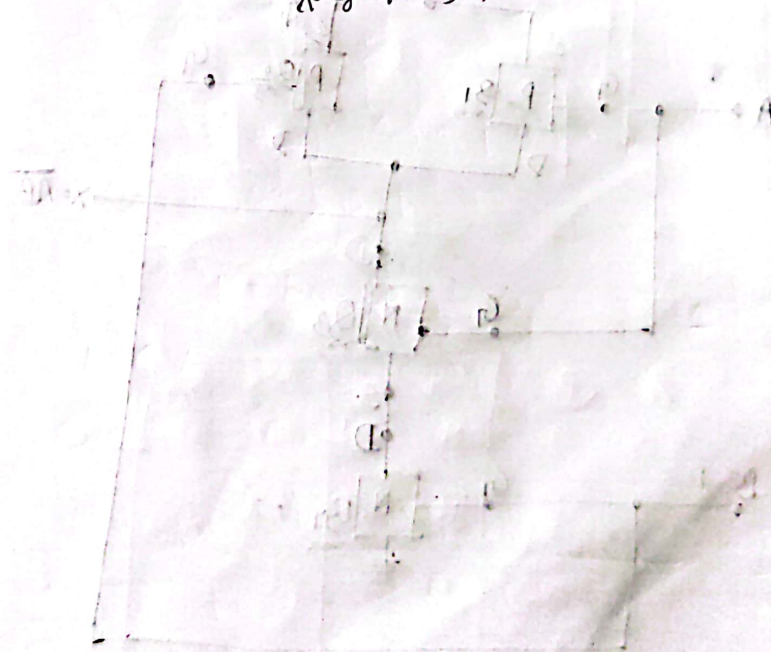
* Noise margin: good noise immunity, they're robust against small variations in input voltage.

* Efficiency: Very widely used in integrated circuits due to low power consumption.

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Applications:

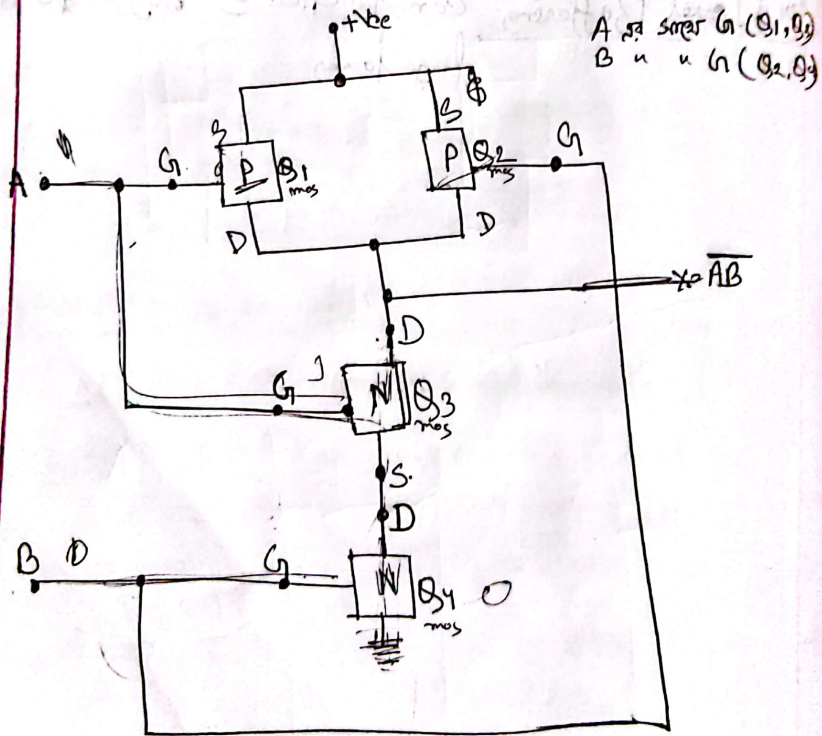
- (i) Used in digital circuitry as NOT gates
- (ii) Memory cells: Used in RAM (SRAM) cells to store binary data.
- (iii) Signal inversion: Used to invert digital signal in circuitry.
- (iv) Level Buffering: can function as buffer to drive large loads.



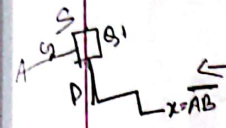
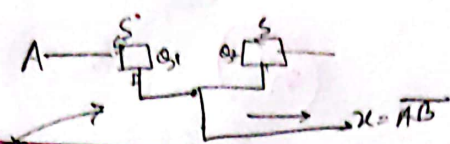
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2MOS NAND:

Input	PMOS	NMOS	
Low	ON	OFF	Normal.
High	OFF	ON	



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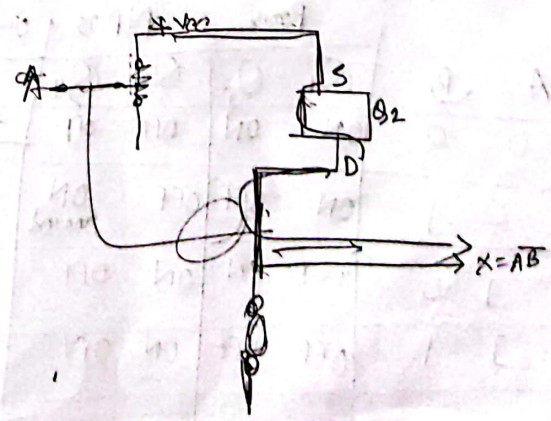


input		PMOS		NMOS		output
A	B	Q1	Q2	Q3	Q4	
0	0	ON	ON	OFF	OFF	1
0	1	ON	OFF	OFF	ON	1
1	0	OFF	ON	ON	OFF	1
1	1	OFF	OFF	ON	ON	0

Short circuit
↓
value ground
9 2015

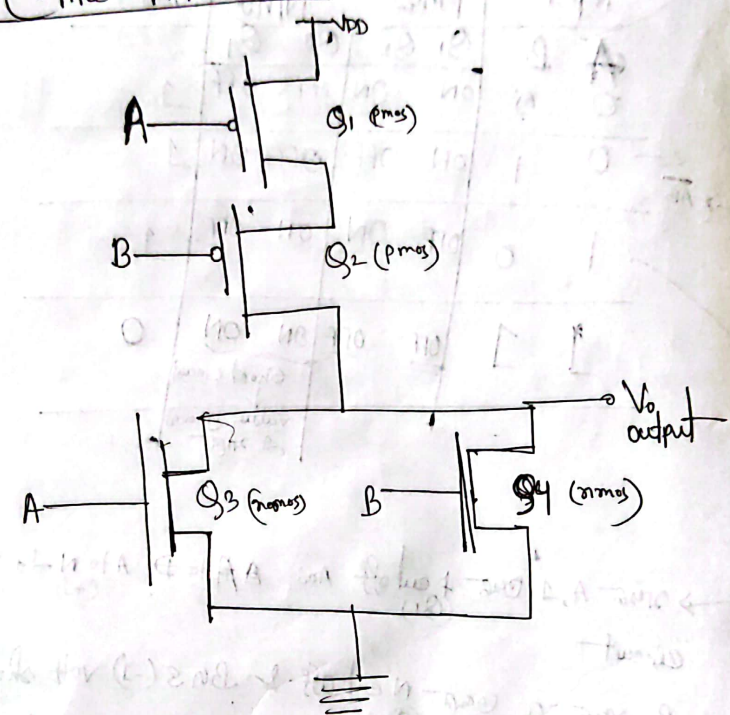
→ when A, 1 then P cut off And A to N-1, short circuit

→ when B, 0 then N cut off & B to S (-1) volt short circuit



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CMOS NOR Gate:



		PMOS		NMOS		output
A	B	Q ₁	Q ₂	Q ₃	Q ₄	
0	0	ON	ON	OFF	OFF	High.
0	1	ON	OFF	OFF	ON ground	LOW
1	0	OFF	ON	ON	OFF	Low
1	1	OFF	OFF	ON	ON	LOW

PTL
V_s July-Dec-2020

Simulation

Noise margin 0:

low output, great high input

$$V_{in} = V_{th} = 0.5, V_o = 0.2 \text{ Silicon}$$

$$\Delta V = (0.5 - 0.2) = 0.3V$$

$$V_{BE} = \frac{I_C}{I_B}$$

$$I_B = 5 \times \frac{I_C}{h_{FE}}$$

Noise margin 1:

high output, low input

$$R_L = 360 \Omega$$

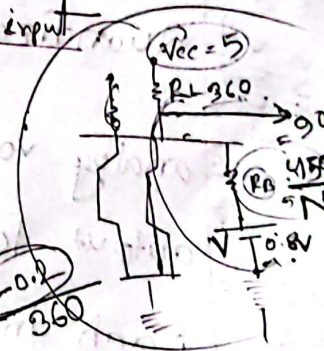
$$R_B = 450 \Omega$$

$$V_{BE} = 0.8$$

$$V_{CE} = 0.2$$

$$N = 5$$

high output, $V_{CC} = 5$



$$V_o = \frac{5 \times 90}{360 + 90} + \frac{360 \times 0.8}{360 + 90}$$

$$= 1.64$$

$$\text{Base current} = N \times I_C$$

$$I_B = 5 \times \frac{0.0133}{10}$$

$$= 0.00667 \text{ mA}$$

$$\left(0.006 \times \frac{450}{N} \right) + 0.8 \xrightarrow{90^\circ} V_{BF}$$

$$= 1.34$$

$$\Delta = 1.64 - 1.34$$

$$= 0.30$$

5 to 20 digi for output